

Daily Task: Rules of Identifiers & Variable Creation

🎯 Task Objective

For each question:

Analyze the Question

Understand the concept

Write the correct answer / explanation

Q1

What is a variable in Python, and how do you create one?

Input: x = 10

Q2

Is it necessary to specify a data type while creating a variable in Python? Why?

Input: x = 10

Input: y = 3.14

Input: z = "Hello"

Q3

How does Python determine the data type of a variable, and can it change later?

Input: x = 10
x = "Python"

Input: value = 5.5
value = 100

Q4

What is variable reassignment, and what happens when a variable is reassigned to a different data type?

Input: a = 25
a = "Data Science"

Input: flag = True
flag = 0

Q5

How can you assign multiple values to multiple variables in a single line?

Input: a, b, c = 1, 2, 3

Input: x, y = "AI", "ML"

Q6

Are variable names in Python case-sensitive? Explain with an example.

Input:
```
value = 10
Value = 20
print(value, Value)
```

Q7

What is an identifier in Python, and what rules must be followed while naming identifiers?

Write your answer

Q8

Can an identifier start with a number or contain special characters? Why?

Input: 1name = "John"

Input: user-name = "Sam"

Q9

Are Python keywords allowed to be used as identifiers? Explain the reason.

Input: def = 10

Input: while = 5

Q10

Why is it important to use meaningful variable names and follow naming conventions in Python?

Input: x = 25

Input: student_age = 25

Input: a = "Python"